

1. Student Management System

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```

private String name;
private String rollNo;
private double[] marks;

```

```

public void setStudentDetails (String name, String rollNo,
                                double[] marks) {

```

```

    this.name = name;
    this.rollNo = rollNo;
    this.marks = marks;

```

```

}

```

2. Library Book Management

```

private boolean isAvailable;
public boolean getAvailability () {
    return this.isAvailable;

```

```

}

```

```

public void issueBook () {
    this.isAvailable = false;

```

```

}

```

3. Employee Salary Calculation

```

private double calculateTax (double gross) {
    return gross * 0.10;

```

```

}

```

```

public double calculateNetSalary () {
    return calculateGrossSalary() - calculateTax (
        calculateGrossSalary());

```

```

}

```

4. Shape Area calculator using Polymorphism

```
public class shape {  
    public double calculateArea() {  
        return 0.0;  
    }  
}  
  
public class circle extends shape {  
    @Override  
    public double calculateArea() {  
        return Math.PI * radius * radius;  
    }  
}
```

5. Vehicle Registration System

```
public class Two Wheeler extends vehicle {  
    public Two Wheeler (String regNo, String owner,  
        String fuelType, int engineCapacity) {  
        super (regNo, owner, fuelType);  
        this.engineCapacity = engineCapacity;  
    }  
}
```

6. Bank Account Hierarchy

```
// Savings Account  
if (balance - amount < 500) {  
    throw new Exception ("Minimum balance limit!");  
}  
  
// current Account  
if (amount <= balance + overdraftLimit) {  
    balance -= amount;  
}
```

7. Hospital patient Record System

```
Private String diagnosis;  
Public String getMedicalRecords (String role) {  
    if (role.equals ("doctor"))  
        return this.diagnosis;  
    return "Access Denied";  
}
```

8. E-commerce Product Catalog

```
Product[] cart = { new Electronics ("E1", 1000),  
                    new clothing ("C1", 500) };  
  
for (Product p : cart) {  
    totalPrice += p.applyDiscount ();  
}
```

9. University Staff Hierarchy

```
Protected double basicSalary;  
Public class Professor extends TeachingStaff {  
    @ override  
    public double calculate Allowance () {  
        return super. calculate Allowance () + 500;  
    }  
}
```

10. Smart Home Device Controller

```
private String secretPIN;
```

```
public void executeCommand(String cmd) {
```

```
    if (cmd.startsWith("unlock:") && cmd.substring(7).  
        equals(secretPIN)) {
```

```
        this.isOpen = true;
```

```
    }
```

```
}
```